

Work Time: 30min

Please copy the subjects and then close your laptops.

Default (1p).

1 (3p). Given the following Java collection:

```
List<Integer> numbers = Arrays.asList(1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 14,15);
```

Using Java functional style (Java streams), please write a Java stream program that is doing the following operations in the following order:

- a)keep the numbers which are multiple of 3 or of 4;
- b)transform each remaining number into its sucesor modulo 3;
- c)compute the sum modulo 2 of the remaining numbers

2 (3p). Given the following four classes in Java:

```
class A extends D{...} class B extends A {...}
```

```
class C extends A{...} class D {...}
```

```
class Amain{
```

```
    D method1(ArrayList<.....> list) { if list.isEmpty() return null; else return list.get(0);}
```

```
    void method2(ArrayList<.....> ) { list.add(null);}
```

```
    void method3(){
```

```
        ArrayList<D> listD=new ArrayList<D>(); listD.add(new A());listD.add(new A());
```

```
        ArrayList<A> listA=new ArrayList<A>(); listA.add(new B());listA.add(new C());
```

```
        ArrayList<B> listB = new ArrayList<B>(); listB.add(new B());listB.add(new B());
```

```
        ArrayList<C> listC = new ArrayList<C>(); listC.add(new C()); listC.add(new C());
```

```
        this.method1(listA); this.method1(listB); this.method1(listC); this.method1(listD);
```

```
        this.method2(listA); this.method2(listB); this.method2(listC); this.method2(listD);
```

```
    }
```

}

Please complete the most specific wildcard types for the class A main methods (method1 and method2) such that the entire program is correct. **Please justify your solution!** If it is not possible to find a solution please explain the reason. **Without justification you will get 0 points!!!**

3 (3p). Is the following Java code correct? Please explain your answer.

```
interface In1 { int getS1();}

class A implements In1 {

    int f1;

    static int s1=0;

    public A(int a) { this.f1=a*s1;s1=s1+1; }

    static int getS() { return getS1(); }

    int getS1() {return s1;}

}
```